Nevo and Rosen (2012): Simulations and Extensions*

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1 Nevo and Rosen (2012)

2 Nevo and Rosen (2012) as Sensitivity Analysis

2.1 Sensitivity Analysis: General Approach

This material has mainly been taken from Paul Diegert's lecture slides on sensitivity analysis.

Consider a family of models indexed by a sensitivity parameter $\delta \in [\underline{\delta}, \bar{\delta}]$:

$$\mathcal{M}(\delta) \subseteq \mathcal{M}(\delta') \subseteq \mathcal{M} \quad \text{for } \delta \leq \delta'.$$

The parameter δ governs the strength of assumptions: larger δ corresponds to weaker assumptions and hence a larger model class.

Assume models are *complete*, i.e. each model $m \in \mathcal{M}$ induces a distribution F^m over observables.

Let F_0 denote the observed data distribution. The identified set under assumption δ is $\mathcal{M}_0(\delta) = \{m \in \mathcal{M}(\delta) : F^m = F_0\}$

For a parameter of interest $\psi : \mathcal{M} \to \mathbb{R}$, the identified set is

$$\Psi_0(\delta) = \{\psi(m) : m \in \mathcal{M}_0(\delta)\}.$$

As δ increases, the model class $\mathcal{M}(\delta)$ expands, so the identified set $\Psi_0(\delta)$ weakly enlarges. Thus δ can be interpreted as a *sensitivity parameter* controlling the degree of identification.

*All code is accessible under https://github.com/mgrotz/NevoRosen2012_SimulationsExtensions

2.2 Connection to Nevo and Rosen (2012)

We can see Nevo and Rosen (2012) imposing the following restrictions with $\delta = 1$:

$$\text{sign}\Big(\text{Cov}(Z,u)\Big) = \text{sign}\Big(\text{Cov}(X,u)\Big), \qquad |\text{Corr}(Z,u)| \leq \delta|\text{Corr}(X,u)|, \qquad \text{Cov}(Z,X) < 0.$$

These restrictions define a set of admissible data-generating processes consistent with observed moments and the structural parameter $\psi(m) = \beta$. This corresponds to specifying a model class $\mathcal{M}(\delta)$, where $\delta \in [0,1]$.

2.3 Identified Set in a Linear Model

2.3.1 Univariate instrument

Redoing calculations from Nevo and Rosen (2012) we find the following bounds:

$$\underline{\beta}(\delta) = \min\left\{\beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)}\right\},$$
$$\overline{\beta}(\delta) = \max\left\{\beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)}\right\}.$$

See Appendix B for the full calculation.

In Figure 1 we plot the identified set $B(\delta) = [\underline{\beta}(\delta), \overline{\beta}(\delta)]$ as a function of δ. As δ increases, the bounds widen, reflecting weaker assumptions and hence less precise identification. We can see that the true β is contained in the identified set for $\delta \geq \frac{\rho_{ZU}}{\rho_{XU}}$, as then Assumption 4 is satisfied.

2.3.2 Bivariate instrument

The identified set with two instruments is just the intersection of the identified sets from each instrument. The fomula is in B. Here we have the following endogeneity assumptions:

$$|\text{Corr}(Z_1,u)| \leq \delta_1|\text{Corr}(X,u)|, \qquad |\text{Corr}(Z_2,u)| \leq \delta_2|\text{Corr}(X,u)|$$

In Figure 2 the dark blue region corresponds to the strength of assumptions we need to make to reject $\beta = 0.4$ in our hypothetical example. The light blue area is the area we are able to reject at now that we have included a second instrument.

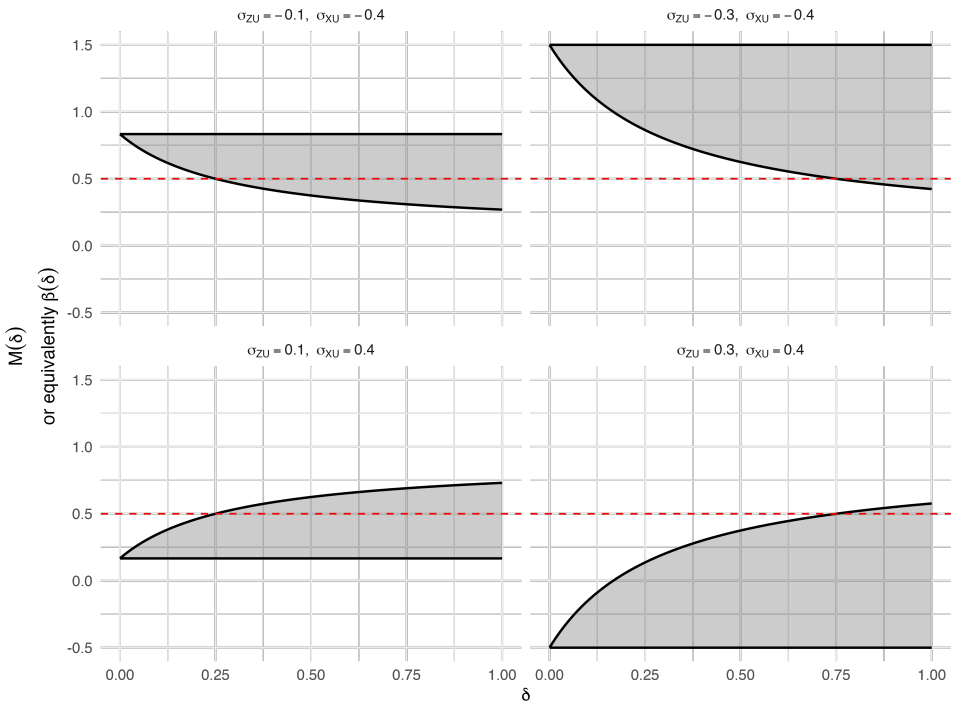

Figure 1: Identified bounds on β as a function of δ. Red line signifies true β. $\sigma_X = \sigma_Z = \sigma_U = 1$ and $\mathrm{Cov}(X, Z) = -0.3$

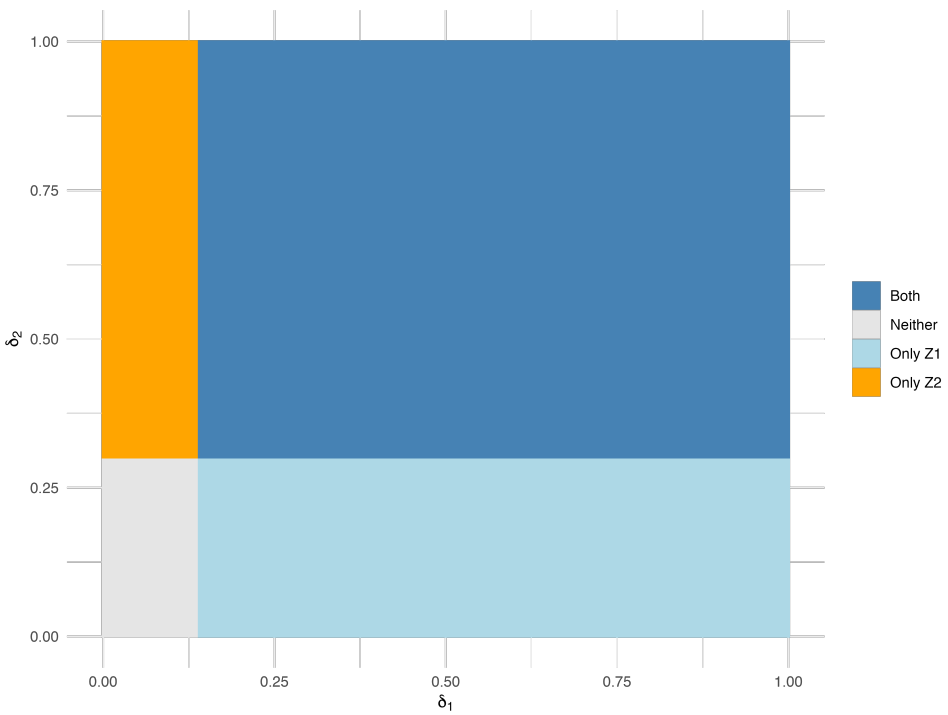

Figure 2: Parameters are set to $\mathrm{Var}(X) = \mathrm{Var}(U) = \mathrm{Var}(Z_1) = \mathrm{Var}(Z_2) = 1$, with $\beta = 0.5$, $\mathrm{Cov}(X, U) = 0.4$, $\mathrm{Cov}(Z_1, U) = 0.1$, $\mathrm{Cov}(Z_2, U) = 0.2$, $\mathrm{Cov}(X, Z_1) = -0.3$, and $\mathrm{Cov}(X, Z_2) = -0.5$. The figure shows, over a grid of (δ_1, δ_2), whether the value 0.4 lies within the identified set based on Z_1 alone, Z_2 alone, and their intersection.

3 General simulations

4 Networks

4.1 Some models

As a recap, the assumptions that we need are:

$$\mathrm{Cov}(Z, X) = 0 \qquad \mathrm{sign}(\mathrm{Cov}(Z, U)) = \mathrm{sign}(\mathrm{Cov}(X, U)) \qquad |\mathrm{Cov}(Z, U)| \leq |\mathrm{Cov}(X, U)|$$

A latent homophily linear-in-means model inspired by can be represented as follows: $y = \beta G y + \gamma x + u, \quad u_i = \rho a_i + \varepsilon_i$. Connections are formed as follows: $A_{ij} = 1\{\eta_{ij} + \kappa_a |a_i - a_j|^{-1} > c\}$ and G is the row-normalized adjacency matrix, so $G = \mathrm{diag}(A\iota_N)^{-1} A$.

Wihtout latent homophily, Bramoullé et al. (2009) suggests using $G^2 x$ as an instrument for Gy. This assumption fails in our model as $\mathbb{E}[(G^2 x)_i u_i] \neq 0$, see Appendix C.

References

Bramoullé, Y., Djebbari, H., & Fortin, B. (2009). Identification of peer effects through social networks. *Journal of Econometrics*, *150*(1), 41–55.

Nevo, A., & Rosen, A. M. (2012). Identification with imperfect instruments. *The Review of Economics and Statistics*, *94*(3), 659–671.

A An Example of Sensitivity Analysis

B Identified Set in Linear Model

B.1 Univariate Instrument

$$Y = \beta X + U, \qquad E[U] = 0.$$

Assume

$$\rho_{XU}\rho_{ZU} \geq 0, \qquad |\rho_{ZU}| \leq \delta|\rho_{XU}|, \qquad \delta \in [0,1].$$

Define

$$\lambda^* = \frac{\rho_{ZU}}{\rho_{XU}}.$$

Then

$$\lambda^* \in [0, \delta].$$

For any $\lambda \in [0, \delta]$, define

$$V(\lambda) = \sigma_X Z - \lambda \sigma_Z X.$$

The IV estimand using $V(\lambda)$ is

$$\beta^{IV}_{V(\lambda)} = \frac{\mathrm{Cov}(V(\lambda), Y)}{\mathrm{Cov}(V(\lambda), X)}.$$

Equivalently,

$$\beta^{IV}_{V(\lambda)} = \frac{\sigma_X \sigma_{ZY} - \lambda \sigma_Z \sigma_{XY}}{\sigma_X \sigma_{XZ} - \lambda \sigma_Z \sigma_X^2}.$$

Hence the identified set is

$$B(\delta) = \left\{ \beta^{IV}_{V(\lambda)} : \lambda \in [0, \delta] \right\}.$$

Since

$$\frac{d}{d\lambda} \beta^{IV}_{V(\lambda)} = \frac{\sigma_X \sigma_Z \left(\sigma_{ZY} \sigma_X^2 - \sigma_{XY} \sigma_{XZ} \right)}{\left(\sigma_X \sigma_{XZ} - \lambda \sigma_Z \sigma_X^2 \right)^2},$$

the sign of the derivative does not depend on λ. Therefore the bounds are attained at the endpoints:

$$\underline{\beta}(\delta) = \min \left\{ \beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)} \right\},$$

$$\overline{\beta}(\delta) = \max \left\{ \beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)} \right\}.$$

Because

$$V(0) = \sigma_X Z,$$

we have

$$\beta^{IV}_{V(0)} = \beta^{IV} = \frac{\sigma_{ZY}}{\sigma_{XZ}}.$$

Thus

$$\boxed{B(\delta) = \left[\min\left\{\frac{\sigma_{ZY}}{\sigma_{XZ}}, \frac{\sigma_X\sigma_{ZY} - \delta\sigma_Z\sigma_{XY}}{\sigma_X\sigma_{XZ} - \delta\sigma_Z\sigma_X^2}\right\},\ \max\left\{\frac{\sigma_{ZY}}{\sigma_{XZ}}, \frac{\sigma_X\sigma_{ZY} - \delta\sigma_Z\sigma_{XY}}{\sigma_X\sigma_{XZ} - \delta\sigma_Z\sigma_X^2}\right\}\right].}$$

B.2 Bivariate Instrument

$$B(\delta_1, \delta_2) = \left[\max\left\{\underline{B}_1(\delta_1), \underline{B}_2(\delta_2)\right\},\ \min\left\{\overline{B}_1(\delta_1), \overline{B}_2(\delta_2)\right\}\right],$$

with

$$\underline{B}_j(\delta_j) = \min\left\{\frac{\sigma_{Z_jY}}{\sigma_{XZ_j}}, \frac{\sigma_X\sigma_{Z_jY} - \delta_j\sigma_{Z_j}\sigma_{XY}}{\sigma_X\sigma_{XZ_j} - \delta_j\sigma_{Z_j}\sigma_X^2}\right\},$$

and

$$\overline{B}_j(\delta_j) = \max\left\{\frac{\sigma_{Z_jY}}{\sigma_{XZ_j}}, \frac{\sigma_X\sigma_{Z_jY} - \delta_j\sigma_{Z_j}\sigma_{XY}}{\sigma_X\sigma_{XZ_j} - \delta_j\sigma_{Z_j}\sigma_X^2}\right\}.$$

C Networks

C.1 Invalidity of standard instrument

To use G^2x as an instrument, we need $\mathbb{E}[(G^2x)_iu_i] = 0$. But in your model, $u_i = \rho a_i + \varepsilon_i$. Therefore, $\mathbb{E}[(G^2x)_iu_i] = \rho\,\mathbb{E}[(G^2x)_ia_i] + \mathbb{E}[(G^2x)_i\varepsilon_i]$. If ε_i is independent of G, x, a, then $\mathbb{E}[(G^2x)_i\varepsilon_i] = 0$. Now write $(G^2x)_i = \sum_{j=1}^n\sum_{k=1}^n G_{ij}G_{jk}x_k$.

Hence

$$\mathbb{E}[(G^2x)_ia_i] = \mathbb{E}\left[a_i\sum_{j=1}^n\sum_{k=1}^n G_{ij}G_{jk}x_k\right].$$

If G is formed from A, and

$$A_{ij} = \mathbf{1}\left\{\eta_{ij} + k_a|a_i - a_j|^{-1} > c\right\}$$

,

then G_{ij} is a function of a_i and a_j. Therefore G^2x is also a function of a_i, through paths $i \to j \to k$.

Thus, generally,

$$\mathbb{E}\left[a_i\sum_{j=1}^n\sum_{k=1}^n G_{ij}G_{jk}x_k\right] \neq 0$$

.

So $\mathbb{E}[(G^2x)_iu_i] \neq 0$.